JJY Antenna for M5Atom

Transmits a simulated JJY standard time signal for radio-controlled clocks







1. Introduction

This printed circuit board is an antenna for transmitting a simulated JJY standard time signal. It sets the time on radio-controlled clocks that are out of reach of the real signal.

2. Requirements

- ✓ An ATOMS3 Lite, ATOM Lite or ATOM Matrix, sold separately (together referred to as “M5Atom”).
- ✓ The sample program is on GitHub.
- ✓ To upload the program, you need a PC that runs the Arduino IDE.
- ✓ Once the program is uploaded, the PC is no longer needed. Power is supplied through USB-C.
- ✓ A permanent internet connection over Wi-Fi.
- ✓ Setting the time takes about 3 to 5 minutes, depending on the clock.

3. Latest information

This manual will go out of date. For the latest information, see:

GitHub: botanicfields/BF-018A https://github.com/botanicfields/BF-018A

4. How to use

(1) Mounting the M5Atom

Mount the M5Atom on the pin headers on the printed circuit board. For safety, check by eye that it is mounted correctly before connecting the power.

(2) Preparing the Arduino IDE

For how to set up the Arduino IDE for the M5Atom, see M5Stack’s website:

“Setting Up the Arduino Development Environment” https://docs.m5stack.com/en/arduino/arduino_ide

Install the following libraries with the Library Manager:

M5Unified, M5GFX, FastLED by Daniel Garcia, WiFiManager by tzapu

With the latest versions of the Arduino IDE, the board package and the libraries, changes in their specifications may stop the program from compiling or working correctly. If that happens, we apologise for the inconvenience, and ask you to use the confirmed versions given on the site in “3. Latest information”.

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(3) Uploading the program to the M5Atom

For the ATOMS3 Lite, download the BF-018ARev4 folder and all the files in it from the site in “3. Latest information” to your PC, then compile BF-018ARev4.ino and upload it.

For the ATOM Lite and ATOM Matrix, please use BF-018ARev3: compile BF-018ARev3.ino and upload it.

(4) Wi-Fi setup

When the M5Atom starts running, after the program is uploaded, after a reset or at power-on, Wi-Fi setup runs first. Wi-Fi setup is done with WiFiManager; for details, see the WiFiManager documentation:

https://github.com/tzapu/WiFiManager

Outline:

① The M5Atom first tries to connect to the access point it connected to last time.

② If it cannot connect, the M5Atom becomes an access point itself and waits for a smartphone or similar device to connect to it.

i. On the smartphone (or other device), connect to the access point the M5Atom has created.

※ SSID: a name based on the chip ID; password/key: none

ii. Open the IP address 192.168.4.1 in a browser.

iii. Enter the SSID/key of the network the M5Atom should connect to.

③ The M5Atom connects to Wi-Fi with the SSID/key you entered.

(5) Transmitting the simulated JJY signal

Once the M5Atom has connected to Wi-Fi and got the date and time by NTP, the LED starts blinking. Place the printed circuit board within 30 cm of the radio-controlled clock, with the face of its spiral pattern (the antenna) towards the clock. Start reception on the clock: setting the date and time takes about 5 minutes. If it does not complete, try changing the position or orientation of the board. It works best in line with the long axis of the bar antenna inside the clock: align the magnetic field from the board with the long axis of the bar antenna, so that the board’s spiral pattern and the bar antenna’s coil are parallel. In good conditions, the time can be set even from about 70 cm away.

Example, placed in line with the long axis of the bar antenna inside the clock, at a distance of 70 cm:

YouTube: BF-018A: JJY Antenna for M5Atom

https://youtu.be/4Soobp9k9r0

(6) LED monitor

The LED is lit while the JJY signal is on. Pressing the button switches the LED monitor on and off. On the ATOM Matrix, only LED 0 is used.

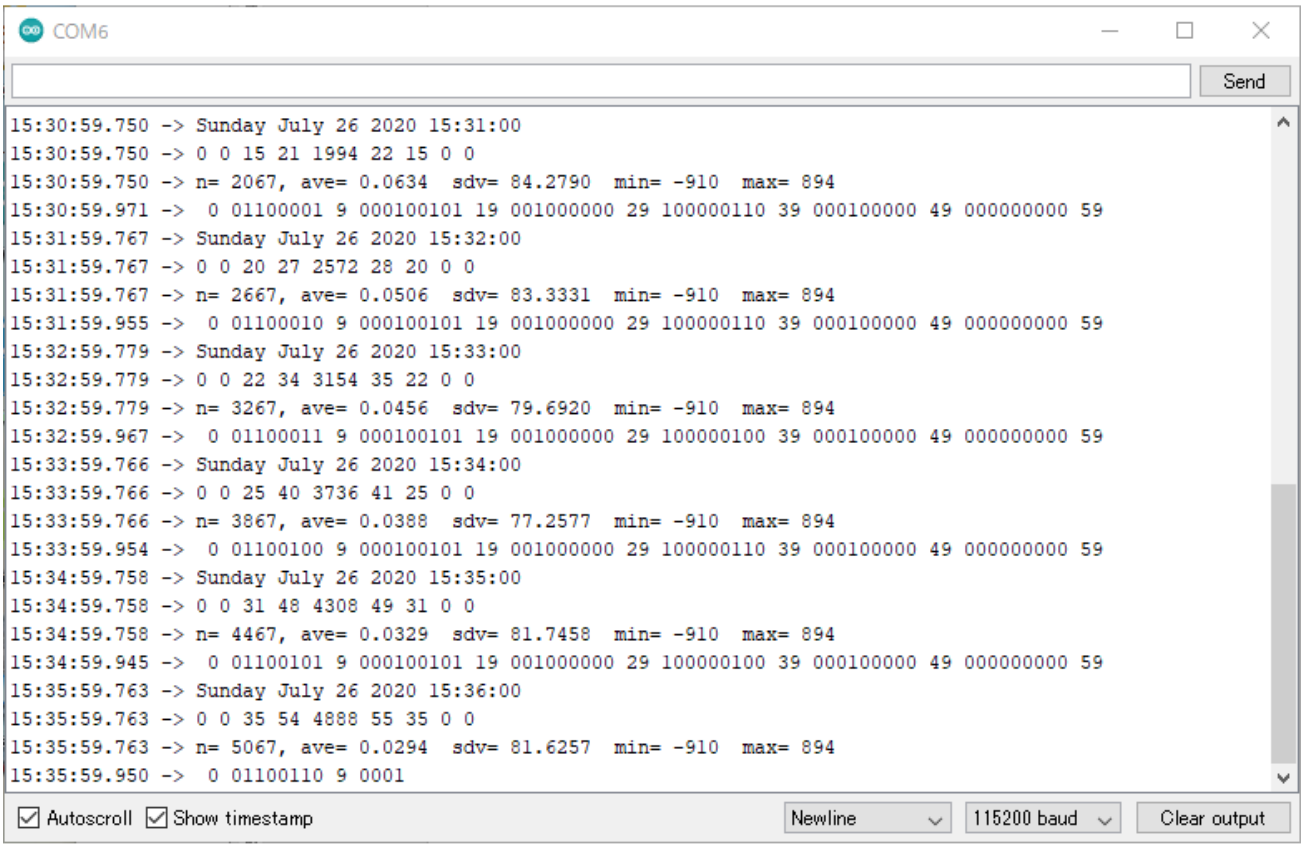
(7) Serial monitor

Log information, such as start-up, the Wi-Fi connection, NTP synchronisation and the generation of the simulated JJY signal, is output to the serial monitor. It helps with troubleshooting. The serial monitor is not required.

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(8) Operation log

An example of the log during operation:



15:30:59.750 ->

- The current time and date
- Distribution of the deviation of the Ticker period (number of occurrences), from the left:

	up to	−50 ms
−50 ms	to	−5 ms
−5 ms	to	−0.5 ms
−0.5 ms	to	−0.05 ms
−0.05 ms	to	0.05 ms
0.05 ms	to	0.5 ms
0.5 ms	to	5 ms
5 ms	to	50 ms
50 ms	and above	

- Number of Ticker periods measured, mean (μs), standard deviation (μs), minimum (μs), maximum (μs)

15:27:32.971 ->

The JJY signal sent:

0 (the first one), 9, 19, 29, 39, 49, 59: markers

Everything else: 0 or 1

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(9) Erasing the Wi-Fi connection settings

Keep the button pressed right after a reset to erase the Wi-Fi connection information stored in the M5Atom. The M5Atom then waits for you to enter the Wi-Fi network to connect to.

(10) RTC

A Unit RTC can be connected to the GROVE port. It is not needed in the usual setup, where the M5Atom is connected to the internet over Wi-Fi. If Wi-Fi cannot connect at power-on or after a reset, the JJY signal can be sent using the date and time kept by the RTC. While the M5Atom is connected to the internet, it updates the RTC with the date and time synchronised by NTP.

References

GitHub: botanicfields/BF-018A
https://github.com/botanicfields/BF-018A

Qiita (in Japanese): Generating a simulated JJY time signal with Ticker on the M5StickC / M5Atom
https://qiita.com/BotanicFields/items/a78c80f947388caf0d36

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V01 26 October 2020

V02 29 May 2021: updated “4. (2) Preparing the Arduino IDE for the M5Atom”

V03 3 December 2022: updated for BF-018ARev2

V04 12 March 2024: updated for BF-018ARev3

V05 9 April 2026: updated for BF-018ARev4

Translator’s note: this is an English translation of the Japanese original, BF-018a_DOC.pdf (V05). The M5Stack link in 4. (2) points to the English version of the page. Corrected in translation: the file name BF-018ARev4.ino in (3), and in (8) the time stamp 15:30:59.750 and the marker at second 9. This repository now also has Rev.5; for its revisions and step-by-step upload instructions, see README.md and FLASHING.md.

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